

BRIELLE JOHNSTON

AI and Data Science Developer

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SUMMARY

AI and Data Science Developer with over six years of experience across predictive modelling and generative AI. Leads the design and delivery of production systems, from model development through to deployed applications, for clients in consulting, cybersecurity, and government.

LANGUAGES: English (Native), French (Native)

EDUCATION

Eastern Michigan University — B.Sc. Actuarial Science

2016 – 2020

- Academic Scholarship
- Athletic Scholarship on the Swim&Dive team
- 3.86/4.0 GPA

TECHNICAL SKILLS

- **AI & Generative AI:** Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, Model Context Protocol (MCP), Semantic Kernel, AutoGen, LangChain, LlamaIndex, and related AI orchestration technologies.
- **Machine Learning & Data Science:** Scikit-learn, PyTorch, Classification, Regression, Time-Series Forecasting, Feature Engineering, Hyperparameter Optimization, Model Evaluation, and related machine learning technologies.
- **Programming Languages & Frameworks:** Python, SQL, TypeScript, FastAPI, Flask, React.
- **Cloud & Data Platforms:** Azure AI Foundry, Azure OpenAI, Azure AI Search, Azure Machine Learning, Cosmos DB, Azure SQL Database, Azure Functions, App Services, Databricks
- **DevOps & Software Engineering:** Docker, Git, CI/CD, REST APIs, Cloud-Native Frontend and Backend Application Development.
- **Certifications:** Databricks Data Engineer Associate; Databricks Machine Learning Associate.

WORK EXPERIENCE

BDO

2023 – Present

SENIOR AI AND DATA SCIENCE DEVELOPER

- Designed and deployed agentic AI solutions using Semantic Kernel, Azure AI Foundry, MCP, n8n, and related agentic AI technologies, enabling multi-agent collaboration, enterprise system integration, dynamic workflow orchestration, and autonomous task execution.
- Designed and implemented production Retrieval-Augmented Generation (RAG) systems used by over 900 employees, using Azure OpenAI, Azure AI Search, FastAPI, and Docker, leveraging prompt optimization, retrieval tuning, automated evaluation workflows and CI/CD-driven deployment processes to deliver citation-grounded enterprise knowledge retrieval solutions.
- Developed intelligent document-processing solutions using OCR, LLMs, and LlamaIndex to automate document generation and transform unstructured content into structured business data, reducing manual processing time from several hours to less than 5 minutes.

- Developed machine learning and deep learning models for classification, regression, and time-series forecasting to predict factory incidents, machinery overloads, and sensor values. Improved model accuracy and reliability through advanced feature engineering, hyperparameter tuning, custom loss functions, and model evaluation techniques, enabling proactive incident prevention and reducing operational downtime.
- Provided technical leadership through architecture reviews, pull request reviews, technology evaluations, and collaborative solution design, helping teams scale AI and data-driven solutions from proof-of-concept to production.
- Conducted technical interviews, mentored junior team members, facilitated knowledge-sharing sessions, and collaborated with stakeholders to identify business requirements and recommend AI-driven technology solutions.

Qohash Inc.

2022 – 2023

DATA SCIENTIST

- Built several machine learning models end-to-end, encompassing data preparation, augmentation, preprocessing, tokenization, model training, hyperparameter tuning, and testing.
- Created a text classifier that provided both predictions and explanations by identifying the words with the greatest influence on each prediction, improving model interpretability for clients.
- Constructed an Optical Character Recognition (OCR) system to extract text from images and documents containing embedded images. Added a model to the OCR pipeline to determine whether an image contained text before processing it.
- Collaborated with software developers to efficiently implement machine learning models into the product.

Privy Council Office

2020 – 2022

DATA SCIENTIST

- Utilized Python and R to preprocess vast datasets and create compelling visualizations to aid executives in decision-making.
- Employed text mining techniques to extract valuable insights from survey text responses and conduct web scraping to seamlessly acquire data from various online sources.
- Created databases, modified them using SQL and used them to increase efficiency.
- Prepared, combined and cleaned large amounts of data in Tableau Prep and Power Queries.
- Created powerful dashboards in Tableau and Power BI.